MILITARY **SPECIFICATION**

RESISTORS, VARIABLE, PRECISION

This specification has been approved by the
Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 Scope.- This specification describes the requirements for the procurement of nine types of precision,variable,wire-wound resistors. Precision Variable Resistor type I, and types III through IX, can be used to obtain an output voltage proportional to applied voltage **and shaft angle of rotation. Precision Variable Resistor** type II can be used to obtatin an output voltage proportional to applied voltage and the tangent of the angle of shaft rotation.

1.2 Classification.- The Precision Variable Resistors shall be of the following types:

Type I	BUWEPS Drawing No. 1989480,	**Function, linear;** total resistance, 1000 ohms; actual electrical angle, 320 degrees
Type II	BUWEPS Drawing No. *1989339,*	Function, tangent (tangent of plus 60 degrees to **tangent of minus 60 degrees); total resistance, 5000 ohms; theoretical electrical angle, 120 degrees each side of center tap**
Type **III**	BUWEPS **Drawing No. 1884975,**	Function, **linear; total resistance,** 100 ohms, **theoretical electrical angle 320 degrees, dual variable resistor**
Type IV	BUWEPS **Drawing No.** 1981272,	Function, linear; total resistance 10,000 ohms; actual electrical angle, *320 degrees; dual* **variable resistor.**

FSC 1290

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Typev	BUWEPS Drawing MO. 1677266,	Function, linear; total resistance 2000 ohms; actual electrical angle, 360 degrees
***Type* VI**	BUWEPS Drawing No. 1989340,	Function, linear; total resistance, 500 ohms; theoretical electrical angle, 360 degrees
Type VII	BUWEPS Drawing No. 1959341	Function, linear; total resistance, 1000 ohms; theoretical electrical angle, 360 degrees; dual variable resistor
Type VIII	BUWEPS Drawing No. 1981398,	Function, linear; total resistance, 1000 ohms; actual electrical angle, 3600 degrees
Type IX	BUWEPS Drawing *NO. 1675104,*	Function, linear; total resistance 10,000 ohms; actual electrical angle, 3600 degrees

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings, and publica- t ions, ***or*** such portion thereof as may be designated herein, of the Issue in effect on the date of invitation of bids, form a part of this specifi- cat ion.

SPECIFICATIONS

FEDERAL

PPP-B-591	Boxes, Fiberboard, Wood-Cleated
PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock-corner
PPP-B-636	Boxes, Fiber
PPP-C-843	Cushioning Material, Cellulosic

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MIL-P-116	Preservation, Methods of

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MIL-S-901	Shockproof Equipment, Class HI (High Impact) Shipboard Application, Tests for
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-P-17555	Preparation for Delivery of Electronic, Electrical and Electromechanical. Equipment and Associated Repair Parts
MIL-F-18870	Fire Control, Optical and Associated Equipment Intended for Shipboard Use, the General Specification for Design and Procurement of
MIL-W-21927	Weapons, Handling, and Preparation for Delivery of; General Requiements for

STANDARDS

MILITARY

MIL-STD-106	Mathematical Symbols
MIL-STD-129	Marking for Shipment and ***Storage***
MIL-STD-167	Mechanical **Vibrations of Shipboard Equipment**
MIL-STD-202	Test Methods for Electronic **and** Electric Component Parts

DRAWINGS

NAVY DEPARTMENT

BUWEPS Dwg 1675104	**Resistor, Variable, Linear, Precision**
BUWEPS Dwg 1677266	**Resistor-Variable, Linear-Precision**
BUWEPS Dwg 1884975	**Resistor, Variable Linear Precision**

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BUWEPS Dwg 1981272	**Resistor,** Variable, Linear, Precision
BUWEPS Dwg 1981398	Resistor, Variable, Linear, Precision
BUWEPS Dwg 1989339	Resistor, Variable, Non-1inear Precision
BUWEPS Dwg 1989340	Resistor, Variable, Linear PrecIsion
BUWEPS Dwg 1989341	Resistor, **Variable, Linear Precision**
BUWEPS Dwg 1989480	**Resistor, Variable, Linear Precision**

PUBLICATRIONS

BUREAU OF NAVAL WEAPONS

NAVWEPS OS 6341	Miscellaneous General ordnance Design Requirements

(Copies of specifications , standards , drawings, and publications requested by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. - The following documents form a part of this specificatIon. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

NATIONAL BUREAU OF STANDARDS PUBLICATION

Handbook H28	Screw-thread Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D. C .)

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules

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(Application for copies should be addressed to the Consolidated Classification Committee, 202 Union Station, Chicago 6, I11.)

AMERICAN TRUCKING ASSOCIATION, INC., PUBLICATION

Motor Freight Classification Rules

(Application for copies should be addressed to the Issuing Officer, American Trucking Association, Inc., 1424 16th Street, N.W., Washington 6, D. C.)

3. REQUIREMENTS

3.1 General requirements-

3.1.1 Pre-production samples. - Unless otherwise specified in the contract or order, pre-production samples of the Precision Variable Resistor shall be manufactured by the methods and procedures proposed for the production lot. These samples shall be tested as specified in Section 4 herein and are for the purpose of determining that the contractor, prior to his starting production, is capable of producing items that comply with the technical requirements of the contract. See 4.1.1.

3.1.2 Materials. - Unless otherwise specified in the contract, requisition, or order, all materials and purchased parts entering into the final assembly of the Recision Variable Resistor shall conform to the detailed requirements given on the applicable BUWEPS drawing and in Specification MIL-F-18870.

3.1.3 Finishes, coatings, construction, and dimensions. - Finishes, coatings, construction, and dimensions for the Recision Variable Resistor shall be in accordance with the applicable drawings, specifications, and publications listed on the applicable BUWEPS drawing.

3.1.4 Moisture, fungus, and corrosion. - The Precision Variable Resistor shall be rendered moisture, fungus, and corrosion resistant in accordance with detail requirements specified on drawings and specifications listed on the applicable BUWEPS drawing.

3.1.5 Marking. - Markings of assemblies shall be in accordance with the applicable BUWEPS drawings, and shall contain the following information unless otherwise specified on the applicable BUWEPS drawing.

(a) BUWEPS drawing number,
(b) Index point as % Rot. = % V. (applicable only to units for which the applicable BUWEPS drawing specifies a terminal conformity requirement),

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(c) **Total resistance in ohms** (to be marked on each resistor cup),
(d) Independent linearity or terminal conformity as specified on the applicable BUWEPS drawing,
(e) Precision Variable Resistor manufacturer's name and part number (for complete unit).

Where insufficient space is available, preference shall be given to the markings in the order in which they are specified In the preceding list .

3.1.6 Index point.

3.1.6.1 Terminal conformity units. - **For Precision Variable Resistors** having a terminal conformity requirement, the following requirements shall be applicable unless otherwise specified on the applicable BUWEPS drawing.

(a The index point **shall** not be either endpoint.
(b) The index point for linear units shall be at 50 percent rotation (midpoint of theoretical electrical angle). The index point for non-linear units shall be on the steep slope portion of the curve.
(c) In the case of ganged units, the index point shall be that of the cup nearest the shaft.

3.1.6.2 Independent linearity units. - For Precision Variable Resistors having an independent linearity requirement, the following requirements shall be applicable unless otherwise specified on the applicable BUWEPS drawing.

(a) The index point shall be the counterclockwise end point.
(b) In the case of ganged units, the index point shall be that of the cup nearest the shaft.

3.1.7 Ganged units. - For Precision Variable Resistors consisting of ganged cups, each cup shall meet the applicable requirements of this specification.

3.1.8 Gages. - The contractor shall provide himself with whatever gages are necessary and adequate to ensure that the material to which this specification applies will meet the requirements shown on the applicable drawings. If so stipulated in the contract, order, or requisition, the Government will furnish drawings of pertinent Navy Final Inspection Gages **for guidance and design of the contractor's inspection gages. However,** such procedure shall not relieve the contractor of his responsibility in the design and manufacture of such gages as may be required for the satisfactory fulfillment of the contract requirements, but is intended to facilitate acceptance of all components and assemblies by Naval Final Inspection Gages.

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3.1.9 **Threads. - Unless** otherwise specified, all threads shall be in accordance with the National Bureau of Standards Handbook H28. The class or fit for threads shall be as specified on the applicable drawings, and the NOT-GO gaging practice shall be as set forth in the aforementioned Handbook H28.

3.1.10 Requests for deviation. - Each **request that a deviation be authorized shall have been submitted to the Procuring Activity, and the deviation shall. have been authorized by the Bureau of Naval Weapons or by a Naval Activity delegated such authority, prior to the production of the affected item, or prior to the performance of a** particular **operation on the item. The deviation shall apply only to this specific item or operation. The request shall define the suggested deviation in sufficient detail to permit the Procuring Activity to evaluate its effect on performance and reliability and the reasons for the suggested deviation shall be stated. Each request that a deviation be authorized, and the authorization, shall state specifically the extent of the departure, the identifying number of the item, the contractor (or bidder) and the contract number. A deviation is essentially temporary in character, and usually represents a departure limited to a specific period, to a specific number of items, or to the contract to which it applies. It may require an equitable reduction in contract price if a decrease in the cost of performance results therefrom.**

3.1.11 **Conflicting** requirements. - conflicting requirements arising between this specification and any specification, publication, or drawing listed herein shall be referred in writing to the procuring agency or appointed agent for interpretation clarification, and resolution (or correction).

3.2 Definitions. - Definitions applicable to this specification and associated drawings are as follows:

3.2.1 Shaft end play. - Shaft end play is **the difference between the high and low dial indicator readings of shaft axial** movement with **a reversing axial load applied to the shaft.**

3.2.2 Shalt radial play. - Shaft radial play is the difference be***tween*** the high and low dial. indicator readings of shaft radial movement with a reversing radial load applied to the shaft.

3.2.3 Shaft rUn-Out. - Shaft run-out is the difference between the high and low dial indicator readings of the shaft at any point on its exposed length when rotated with the housing held stationary.

3.2.4 Pilot run-out. - Pilot run-out is the difference between the high and low dial indicator readings of mounting pilot radial movement with the shaft held stationary and the housing rotated.

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3.2.5 Lateral run-out. - Lateral run-out is the difference between the high and low dial indicator readings of mounting surface axial movement with the shaft held stationary and the housing rotated.

3.2.6 Torques.

3.2.6.1 starting torque. - Starting torque is the maximum torque required to start the shaft rotating when the applied torque is increased slowly from zero.

3.2.6.2 Running **torque. - Running torque is the maximum torque required to maintain continuous, smooth shaft rotation when the applied torque is increased slowly from zero.**

3.2.6.3 stop torque. - Stop torque is the static torque which can be applied to the mechanical stops without resulting damage.

3.2.7 Angles.

3.2.7.1 Mechanical angle. - Mechanical angle is the angle through which the shaft is rotated in going from either Mechanical stop to the other. When there are no mechanical stops, continuous rotation is possible.

3.2.7.2 Continuity angle. - Continuity angle is the greatest angle through which the shaft can be rotated while electrical continuity is maintained between the wiper and each input terminal.

3.2.7.3 Actual electrical angle. - Actual electrical angle is the angle through which the shaft is rotated in going from either end point to the other.

3.2.7.4 Theoretical electrical angle. - Theoretical electrical angle is a specified increment of shaft angle through which it is required that the terminal. conformity requirements of the applicable BUWEPS drawing be met .

3.2.8 Index point. - Preference shaft angular position with respect to which shaft angle can be measured. For units having an independent linearity requirement, the index point is the shaft position for **which** the wiper is at the counterclockwise end point. For units having a terminal conformity requirement, the index POints is the shaft position for which the ratio of wiper voltage to input voltage is equal to the ratio marked on the unit by the manufacturer.

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3.2.9 End point. - An end point is a point where the rate of change of wiper to nearest input terminal resistance with respect to shaft angle changes abruptly as the wiper moves from the overtravel region toward the resistance element. When the Mechanical angle is less than the angle encompassed by the resistance element, the end points are the Mechanical stop positions.

3.2.10 Independent linearity. - Independent linearity is the maximum deviation of actual output voltage from a theoretical output voltage defined by a straight line drawn through the actual output versus rotation curve over the entire actual electrical angle, with the slope and position of the ***straight line*** chosen to minimize the maximum deviation. Independent linearity is expressed as a percentage in volts maximum deviation per input volt.

3.2.11 Terminal conformity. - Terminal conformity is the maximum deviation of actual output voltage from a theoretical output voltage defined by a straight line drawn through the points zero output voltage at the lower limit of theoretical electrical angle and output voltage equals input voltage at the upper limit of theoretical electrical angle. Terminal conformity is expressed as a percentage in volts maximum deviation per input volt.

.3.2.2 Tap. -A tap is an electrical connection made to the resistance element at any point between the end points. Tap location is defined as being at the mechanical center of the effective tap width.

3.2.12.1 Effective tap width. - Effective tap width is the angle of mechanical rotation throughout which there is no change in voltage or resistance as the wiper is moved past the tap in either direction.

3.2.13 Electrical resolution. - Electrical resolution is a measure of accuracy to Which the Precision Variable Resistor output voltage may be set with reference to the input voltage. The shaft is moved the smallest measurable displacement angle which will produce a finite change in output voltage. The maximum incremental change in output voltage thus measured, observed anywhere in the mechanical shaft rotation, when multiplied by 100 and divided by the input voltage is the percentage electrical resolution.

3.2.14 Tropical environment. - Tropical environment is defined as conditions of high humidity, high temperatures, and spore exposure, which present an opportunity for fungi growth.

3.3 Performance requirements and product characteristics. - The Recision Variable Resistor shall meet the following performance requirements and product characteristics:

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3.3.1 Mechanical **requirements .**

3.3.1.1 Shaft end play. - When measured in accordance with 4.4.4.1, the shaft end play shall not exceed 0.005 inch total indicator reading **(T.I.R.) unless otherwise specified on the applicable BUWEPS drawing.**

3.3.1.2 Shaft radial play. - When measured in accordance with 4.4.4.2, the shaft radial play shall not exceed 0.001 inch T.I.R. unless otherwise specified on the applicable BUWEPS drawing.

3.3.1.3 shaft run-out. - **When measured** in **accordance with** 4.4.4.3, the shaft run-out shall meet the requirements of the applicable BUWEPS drawing.

3.3.1.4 Pilot run-out. - When measured in accordance with 4.4.4.4, the pilot run-out shall meet the requirements of the applicable BUWEPS drawing.

3.3.1.5 ***Lateral run-out.*** **- When measured in accordance with 4.4.4.5, the** lateral run-out of the mounting face shall meet the requirements of the applicable BUWEPS drawing.

3.3.1.6 starting torque. - When measured in accordance with 4.4.4.6, the starting torque shall meet the requirements of the applicable BUWEPS drawing.

3.3.1.7 Running torque. - When measured in accordance with 4.4.4.7, the running torque of Precision Variable Resistor types III through IX shall meet the requirements of the applicable BUWEPS drawing.

3.3.1.8 Mechanical angle. - When checked in accordance with 4.4.4.8, the mechanical angle - limited or continuous, as applicable - for Precision Variable Resistor types II through IX shall meet the requirements of the applicable BUWEPS drawing.

3.3.1.9 Continuity angle. - When measured in accordance with 4.4.4.9, the continuity angle of Recision Variable Resistor types I, II, IV, V, and IX shall meet the requirements of the applicable BUWEPS drawing.

3.3.1.10 Stop torque. - When Precision Variable Resistor types I, VIII, and IX are tested in accordance with 4.4.4.10, following application to each mechanical stop of shaft torque equal to the lower limit of stop torque specified on the applicable BUWEPS drawing, the following conditions shall be satisfied.

(a) Starting torque shall meet the requirements of the applicable BUwEPS drawing*

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(b) The change in mechanical angle **from the value measured prior " to the stop torque test shall not exceed 1 degree.**

(c) The continuity angle shall be equal to the mechaical angle for Precision Variable Resistor types I and VIII, and shall meet the requirements of BuWEPS drawing 1675104 for Precision Variable Resistor type IX. See 4.5.4.10.

3.3.2 Electrical requirements.

3.3.2.1 Dielectric strength and insulation resistance.

3.3.2.1.1 Dielectric strength. - When subjected to high-potential voltage in accordance with 4.4.4.11.1, the Precision Variable Resistor shall withstand the voltage without damage, arcing, or breakdown, and the current shall not exceed 3 milliamperes unless otherwise specified on the applicable BUWEPS drawing.

3.3.2.1.2 Insulation resistance. - When measured in accordance with *4.4.4.11.2,* the insulation resistance between ***all terminals*** tied together and the shaft shall be 100 megohms minimum.

3.3.2.2 Total resistance. - When measured in accordance with *4.4.4.12,* the total resistance shall meet the requirements of the applicable BUWEPS drawing.

3.3.2.3 Actual electrical angle. - When measured in accordance with 4.4.4.13, the actual electrical angle of Precision Variable Resistor types 1, IV, V, VIII, and IX shall meet the requirements of the applicable BUWEPS drawing.

3.30204 Wiper phasing. - When measured In accordance with 4.4.4.14, the angle between the two wipers of Precision Variable Resistor type VII shall meet the requirements of BUWEPS drawing 1989341.

3.3.2.5 Tap angle. - When measured in accordance with 4.4.4.15, the tap **position or the angle between taps, as applicable, for Precision Variable** Resistor types II, IV, V, and VI shall meet the requirements of the applicable BUWEPS drawing, and the angle between taps for Precision Variable Resistor type VII shall be 180 degrees plus or minus 1 degree.

3.3.2.6 **Tap width. - When measured in accordance With 4.4.4.16, the width of each tap of Precision Variable Resistor types II, IV, V, VI, and VII shall be 1 degree maximum unless otherwise specified on the ap-**plicable BUWEPS drawing.

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3.3.2.7 Electrical resolution. - When measured in accordance with 4.4.4.17, the electrical resolution shall meet the requirements of the applicable BUWEPS drawing.

3.3.2.8 Independent linearity. - When measured in accordance with *4.4.4.18,* the independent linearity of Precision Variable Resistor types 1, IV, V, VIII, and IX shall meet the requirements of the applicable BUWEPS drawing.

3.3.2.9 Terminal conformity. - When measured in accordance with 4.4.4.19, the terminal conformity of Precision Variable Resistor types II, III, VI, and VII shall meet the requirements of the applicable BUWEPS drawing.

3.3.2.10 Noise. - When measured in accordance with *4.4.4.20,* the electrical noise shall not exceed 100 millivolts peak to peak.

3.4 Environmental requirements. - The Precision Variable Resistor shall meet the other applicable requirements of this specification during or after tests defined or specified in 4.5.

3.4.1 Vibration, operating condition. - The Precision Variable Resistor shall withstand the Type 1 vibration requirements specified in Standard MIL-STD-167, shall function during and after vibration tests, and thereafter shall show no damage and shall meet all the performance requirements and product characteristics of this specification.

3.4.2 Vibration-transportation. - The Precision Variable Resistor, after being subjected to the tests specified in 4.5.2, shall show no damage and shall meet all the requirements of' SectIon *3.*

3.4.3 Shock, non-operating condition. - The Precision Variable Resistor shall withstand the Type A shock requirements specified in Specification MIL-S-901, and thereafter shall show no damage and shall meet all the performance requirements and product characteristics of this specification.

3.4.4 Shock-transportation, storage and handling. -The ***Recision*** Variable Resistor, after being subjected to the tests specified in 4.5.4, shall show no damage and shall meet all the requirements of Section 3.

3.4.5 Temperature, operating condition. - The Precision Variable Resistor shall withstand operational ambient temperatures between the limits of the ambient **temperature *range*** specified on the **applicable** BUWEPS drawing **in accordance** with Specification MIL-F-18870, shall function during **and** after temperature tests, and thereafter shall show no damage or corrosion and shall meet all the performance requirements and product characteristics of. this specification.

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3.4.6 Temperature, non-operating condition. - The Precision Variable Resistor shall withstand storage temperatures Of between minus 67 degrees Fahrenheit (minus 55 degrees centigrade) and the upper limit of ambient temperature range specified on the applicable BUWEPS drawing in accordance with Condition II of OS 6341, and thereafter shall show no corrosion or damage and shall meet all the performance requirements and product characteristics of this specification.

3.4.7 Humidity. - The Precision Variable Resistor shall withstand operational ambient humidities in accordance with Specification MIL-F-18870, and thereafter shall show no corrosion or damage and shall meet all the performance requirements and product characteristics of this specification.

3.4.8 Tropical environment. - The Precision Variable Resistor shall withstand the tropical environmental conditions specified in Specification MIL-E-5272 under "Fungus Resistance Tests", and thereafter shall show no fungus growth, corrosion, or other damage, and shall meet all the performance requirements and product characteristics of this specification.

3.4.9 **Operating life.**

3.4.9.1 100,000 **cycles. - The Precision** Variable Resistor shall withstand 100,000 cycles of **operation** under the conditions specified in 4.5.9.1, and thereafter shall meet all the performance requirements and product characteristics of this specification except that:

(a) A starting torque not exceeding 150 percent of the starting torque specified on the **applicable BUWEPS drawing shall be acceptable,**

(b) An electrical noise voltage not exceeding 400 millivolts peak to peak shall be acceptable.

3.4.9.2 200,000 cycles. - The Precision Variable Resistor shall withstand 200,000 cycles of operation under the conditions specified in 4.5.9.2, and thereafter shall meet all the performance requirements and product characteristics of this specification except that:

(a) A starting torque not exceeding 200 percent of the starting torque specified on the applicable BUWEPS drawing shall be acceptable

(b) An independent linearity or terminal conformity, as applic - able, not greater than 150 percent of the value specified on the applicable BUWEPS drawing shall be acceptable.

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(c) An electrical noise voltage not exceeding 400 millivolts peak to peak shall be acceptable.

3.5 Clean-up. - Prior to and after final assembly, all parts, components and the assembly shall be thoroughly cleaned of loose, spattered, or excess solder, metal chips, and other foreign matter. Burrs and sharp edges , as well as rosin flash, shall be removed.

3.6 Workmanship. - The equipment, including all parts **and accessories, shall be** constructed and finished in a manner to assure compliance with all the requirements of this specification. Particular attention shall be paid to neatness and thoroughness of soldering, marking of parts and assemblies, riveting, machine screw assemblage, and freedom of parts from burrs and sharp edges.

3.6.1 The workmanship exhibited in the pre-production sample will be evaluated to determine acceptability. The approved standards of workmanship will thereby become a minimum production requirement for all units offered for acceptance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified here in. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Pre-product ion sample. - Unless otherwise specified for each contract or order, a representative sample of three Precision Variable Re - sistors shall be manufactured by the methods and procedures proposed for the production lot. These shall be submitted for pre-production tests at an activity designated **by the** Bureau of Naval Weapons, for purposes of determining compliance with the requirements of the contract, specifications, and drawings. Further production of the item by the contractor prior to approval of the contracting activity or the completion of tests on the pre - production samples shall be at the contractor's risk. Pre-production samples accepted will be applied as part of the quantity specified by the contract or order, but shall not be used in any equipment manufactured for the Government.

4.1.2 Periodic production samples. - Tests of 4.4 and *4.5* shall be performed in steps, as specified below. Periodic production samples ac - cepted will be applied as part of the quantity specified by the contract or order, but shall not be used in any equipment manufactured for the government.

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step 1. - **Upon** initiation of production, select one (1) unit from each group of ten (10) units for submission to an activity designated by the procuring activity.

Step 2. - Upon **satisfactory compliance of each unit from three (3) consecutive groups, select one (1) unit from each group of twenty-**five (25) units.

step 3. - Upon **satisfactory compliance of each unit from three (3) cousecutive groups, select one (1) unit from each group of fifty** (50) units. Continue this procedure until a defective unit is found.

Step 4. - When a defective unit is found in any group, the group represented by the sample shall be rejected. Shift to step 1 **and proceed as indicated. Disposition of groups represented by failing samples shall be made by the Government.**

4.2 Classification of tests. - The inspection and testing of the Recision Variable Resistor shall be classified as follows:

4.2.1 Acceptance tests. - These tests shall be accomplished on each Precision Variable Resistor submitted for acceptance under contract. These tests are detailed in 4.4.

4.2.2 Pre-production and periodic production tests. -Pre-production and periodic production tests are those which are accomplished on samples selected as specified in 4.1.1 and 4.1.2. These tests are detailed in 4.5. Acceptance will be based on no defects in a sample.

4.2.3 Tests after delivery. - Unless otherwise specified in the contract, the Government reserves the right to conduct any or all the tests specified herein, as deemed necessary, after installation of the Recision Variable Resistor. Any defects discovered during or after these tests due to the contractor's failure to comply with drawings or specifications for this equipment shall be corrected by the contractor at no expense to the Government, provided the contractor is notified of such defects within the guarantee period of the contract.

4.3 Test equipment. - The following items of test equipment are required to perform the acceptance tests set forth in this specification.

4.3.1 End play tester. - An end play tester or bench lathe capable of securing the Preision Variable Resistor in a vertical or horizontal position and mounting a dial indicator.

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4.3.2 Dial indicator. - A dial indicator which can be mounted On the tester of 4.3.1, and which measures with an accuracy of plus or minus 0.0001 inch.

4.3.3 Mechanical loads*

4.3.3.1 Reversing l/2-pound axial load. - A *reversing* l/2-pound load which can be applied axially to the shaft of the Precision Variable Resistor.

4.5.5.2 Reversing l/2-pound radial load. - A reversing l/2-pound load Which can be applied radially to the shaft of the Precision Variable Resis - tor.

4.3.4 Torque gages.

4.3.4.1 0.1 ounce-inch torque wrench . - A torque wrench capable of applying and measuring a torque of 0.1 ounce-inch with an accuracy of plus or minus 5 percent.

4.3.4.2 2.0 ounce-inch torque wrench. - A torque wrench capable of applying and measuring a torque of 2.0 ounce inches with an accuracy of plus or minus 5 percent .

4.3.4.3 20 pound-inch torque wrench. - A torque wrench capable of applying and measuring a torque of 20 pound-inches with an accuracy of plus or minus 5 percent.

4.3.5 Angle divider. - An angle divider having continuous 360-degree rotation with a maximum angular error of plus or minus 20 seconds of arc. The angular *error shall* include all positioning, coupling, and reading errors.

4.3.6 Ohmmeter. - A multi-range ohmmeter having mid-scale deflections of approximately 15, 1500, and 150,000 ohms, and a mid-scale accuracy of plus or minus 5 percent. The maximum current the ohmmeter applies on any range shall not exceed 10 milliamperes.

4.3.7 Whetstone bridge. - A direct-current resistance bridge capable of measuring resistance from 300 to 30,000 ohms with an accuracy of plus or minus 0.3 percent. The maximum current the bridge applies on any range shall not exceed 1 milliampere.

4.3.8 High potential tester. - A high potential tester capable of applying up to 1000 volts, 60 cycles per seconds which cuts out or indi- cates when the load current exceeds 3 milliamperes. The rate of change of output voltage shall not exceed 75 volts per second when the tester is turned on or Off.

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4.3.9 Insulation resistance tester. - An insulation tester capable of applying 500 volts direct current, and measuring resistance up to **200 mgohms.**

4.3.10 Power supply. - A direct -current power supply having the following characteristics:

(a) Voltage. - Continuously adjustable from 3 to 100 volts,
(b) Current. - 100 milliamperes minimum,
(c) Regulation. - The output voltage shall not change by more than 0.5 volt for a 100-milliampere change in load current.

4.3.11 Voltmeter. - A multi-range, direct-current voltmeter, approximately 0.3, 1, 3, 10, 30, and 100volts full scale, having an input impedance of 20,000 ohms per volt minimum, and an accuracy of plus or minus 3 percent of full scale.

4.3.12 Decade voltage divider. - An adjustable resistance voltage divider, having an accuracy of output-to-input ratio of plus or minus 0.01 percent, an input resistance of 10,000 ohms, and adjustable in steps of 0.01 percent.

4.3.13 Null detector. - Either of the following shall be used.

4.3.13.1 Micrometer. - A direct-current electronic microammeter, 5 microamperes full. scale, having a resistance of 250 ohms maximum, and an accuracy of plus or minus 5 percent of full scale.

4.3.13.2 Millivoltmeter. - A direct-current electronic millivoltmeter, 10 millivolts full scale, having an input resistance of 25,000 ohms minimum-, and an accuracy of plus or minus 5 percent of full scale.

4.3.14 oscilloscope. - A calibrated, direct-coupled oscilloscope having a minimum input impedance of 250,000 ohms, a minimum frequency response of plus or minus 3 decibels from direct current to 50,000 cycles per second, and a minimum sensitivity of 100 millivolts peak to peak **per inch.**

4.3.15 **Decade** resistor. - A decade resistor, adjustable from 10 to 100,000 ohms in steps of 10 ohms, having an accuracy of plus or minus 5 percent, and a current rating of 1 milliampere minimum.

4.4 Acceptance tests.

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4.4. 1 **The** contractor shall conduct acceptance as specified in 4.2.1 to assure that the Precision Variable Resistor is in compliance with the requirements of 3.1, 3.3, 3.5, and 3.6.

4.4.2 **Test** conditions. - Unless otherwise specified, the Precision Variable Resistor shall be subjected to acceptance tests under the following conditions:

4.4.2.1 Temperature: Room ambient 65 to 95 degrees Fahrenheit (18 to 35 degrees centigrade).

4.4.2.2 Altitude: Normal ground atmospheric pressure.

4.4.2.3 Vibration: None.

4.4.2.4 Humidity: Room ambient to 95 percent relative maximum.

4.4.3 **Test and inspection** equipment and facilities. - The manufacturer shall furnish and maintain all necessary test equipment, facilities , and personnel for performing all acceptance tests. The test equipment shall be adequate in quantity and, unless definite requirements are specified, shall be of sufficient accuracy and quality to permit performance of the required acceptance tests.

4.4.4 Test procedure. - The detailed test procedure listed in the following paragraphs shall be followed.

4.4.4.1 Shaft end play. - With the Precision Variable Resistor mounted horizontally in the tester of 4.3.1, the housing held stationary, and the indicator of 4.3.2 mounted to measure axial movement of the shaft, the re - versing l/2-pound load of 4.3.3.1 shall be applied axially to the shaft, and the total indicator reading shall be observed to determine compliance with the requirements of 3.3.1.1.

4.4.4.2 **Shaft** radial play. - With the Precision Variable Resistor mounted horizontally in the tester of 4.3.1, the housing held stationary, and the indicator of 4.3.2 mounted to measure radial movement of the shaft as close as possible to the face of the bearing, the reversing l/2-pound load of 4.3.3.2 shall be applied radially to the shaft approximately 1/2 inch from the face of the bearing, and the total indicator reading shall be observed to determine compliance with the requirements of 3.3.1.2.

4.4.4.3 Shaft run-out. -With the Precision Variable Resistor mounted horizontally in the tester of 4.3.1, the housing held stationary, and the Indicator of 4.3.2 mounted to measure radial movement of the shaft 1/16 inch from the end of the shaft, the shaft shall be rotated, and the total indicator reading shall be observed to determine compliance with the requirements of 3.3.1.3.

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4.4.4.4 Pilot run-out. - With the Pecision Variable Resistor mounted vertically in the tester of 4.3.1, the shaft held stationary, and the indicator of 4.3.2 mounted to ***measure*** radial movement of the pilot, the housing shall be rotated, and the total indicator reading shall be observed to determine compliance with the requirements of 3.3.1.4.

4.4.4.5 **Lateral inn-out. - With the Precision** Variable resistor mounted vertically in the tester of 4.3.1, the shaft held stationary, and the indicator of 4.3.2 mounted to measure axial movement of the mounting face not more than 0.062 inch from its outside diameter, the housing shall be rotated, and the total indicator reading shall be observed to determine compliance with the requirements of 3.3.1.5.

4.4.4.6 starting torque. - **With the torque gage of** 4.3.4.1 or 4.3.4.2, as applicable, attached to the shaft of the Precision Variable Resistor and held stationary, torque shall be applied by hand to the housing and increased slowly until the housing starts to rotate. This operation shall be repeated for several different wiper positions for both directions of rotation. The greatest value of torque required to start rotation of the housing shall be noted to determine compliance with the requirements of 3.3.1.6.

4.4.4.7 Running torque. - With the torque-wrench of 4.3.4.1 attached to the shaft of the Precision Variable Resistor and held stationary, the housing shall be rotated by hand at an angular rate great enough to permit continuous, smooth rotation. This procedure shall. be repeated for several different wiper positions for both directions of rotation. The greatest value of torque required to maintain continuous smooth rotation of the housing shall be noted to determine compliance with the requirements of 3.3.1.7.

4.4.4.8 Mechanical angle. - The angle **between mechanical stops shall be measured using the angle divider of** *4.3.5* **or continuity of shaft** rotation shall be checked manually, as applicable, for Recision Variable Resistor types II through IX to determine compliance with the requirements of 3.3.1.8, and the angle between mechanical stops for Recision Variable Resistor type I shall be measured, using the angle divider of 4.3.5, for reference.

4.4.4.9 Continuity angle. - The continuity angle of Precision Variable Resistor types I, II, IV, and IX shall be measured using the following procedure to determine compliance with the requirements of 3.3.2.3. Using the angle divider of 4.3.5, and with the ohmmeter of 4.3.6 connected between the wiper and either input terminal, the shaft shall be rotated slowly ***over*** the entire mechanical angle - limited or continuous, as applicable - while observing the ohmmeter reading, and the procedure shall then be repeated for the ohmmeter connected between the wiper and the other input terminal.

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4.4.4.10 Stop torque. - For Precision Variable Resistor types I, VIII, and IX, the shaft shall be turned to bring the wiper into contact with either mechanical stop, and the wiper shall be forced against the mechanical stop by applying shaft torque equal to the lower limit of stop torque specified on the applicable BUWEPS drawing, using the torque wrench of 4.3.4.3. The shaft shall then be turned to bring the wiper into contact with the other mechanical stop, and the same value of torque shall be applied.

Following application of stop torque to the two mechanical stops, the **following procedure shall be followed.**

(a) The starting torque of Precision Variable Resistor types I, VIII; and IX shall be measured using the method specified in 4.4.4.6 to determine compliance with the requirements of 3.3.l.10(a).

(b) The angle between mechanicsl **stops for precision Variable Resistors types I, VIII, and IX shall be measured using the angle divider of** *4.3.5,* **and compared with the corresponding values of mechanical angle measured in** *4.4.4.8,* to **determine compliance** with the requirements of 3.3.l.10(b).

(c) Continuity angle **shall be** measured for precision Variable Resistor types 1, VIII, and IX using the method specified in 4.4.4.9 to determine compliance with the requirements of 3.3.1.1O(C).

4.4.4.11 Dielectric strength and insulation resistance.

4.4.4.11.1 Dielectric strength. - Using the high potential tester of 4.3.8, the dielectric test voltage specified on the applicable BUWEPS drawing, or 750 volts r.m.s. for precisionVariable Resistor type III, as applicable, shall be applied between all terminals tied together and the shaft for 1 minute unless otherwise specified on the applicable BUWEPS drawing. The high potential tester shall be observed for indication of current during application of high potential voltage, and the precision Variable Resistor shall. be examined visually after application of high potential voltage, to determine compliance with the requirements of 3.3.2.1.1.

4.4.4.11.2 Insulation resistance. -Using the insulation resistance tester of 4.3.9, the insulation resistance between all terminals tied together and the shaft shall be measured, applying the insulation resistance tester direct-current voltage for 1 minute, to determine compliance with the requirements of 3.3.2.1.1.

4.4.4.12 **Total resistance. - The total resistance shall be measured using the Whetstone bridge of** 4.3.7 to determine compliance with the re**quirements of 3.3.2.2.**

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4.4.4.13 Actual electrical angle. - With Precision Variable Resistor type I, IV, V, VIII, or IX mounted in the angle divider of 4.3.5, and the ohmmeter of 4.3.6 connected between the wiper and either input terminal, the shaft shall be rotated slowly over the entire mechanical angle - limited or continuous, as applicable - while observing the ohmmeter indication, and the procedure shall then be repeated for the ohmmeter connected between the wiper and the other input terminal, to determine compliance with the requirements of 3.3.2.3.

4.4.4.14 Wiper phasing. - With Precision Variable Resistor type VII connected in accordance with figure 1, and the null detector connected between terminals 1 and 2 of cup A, the shaft shall be positioned to obtain a null indication, and this shaft angle shall be noted. This procedure shall be repeated for the null detector connected between terminals 2 and 3 of cup B, and the difference of the two shaft angles (the wiper phasing error) shall be computed to ***determine compliance*** with the requirements of 3.3.2.4.

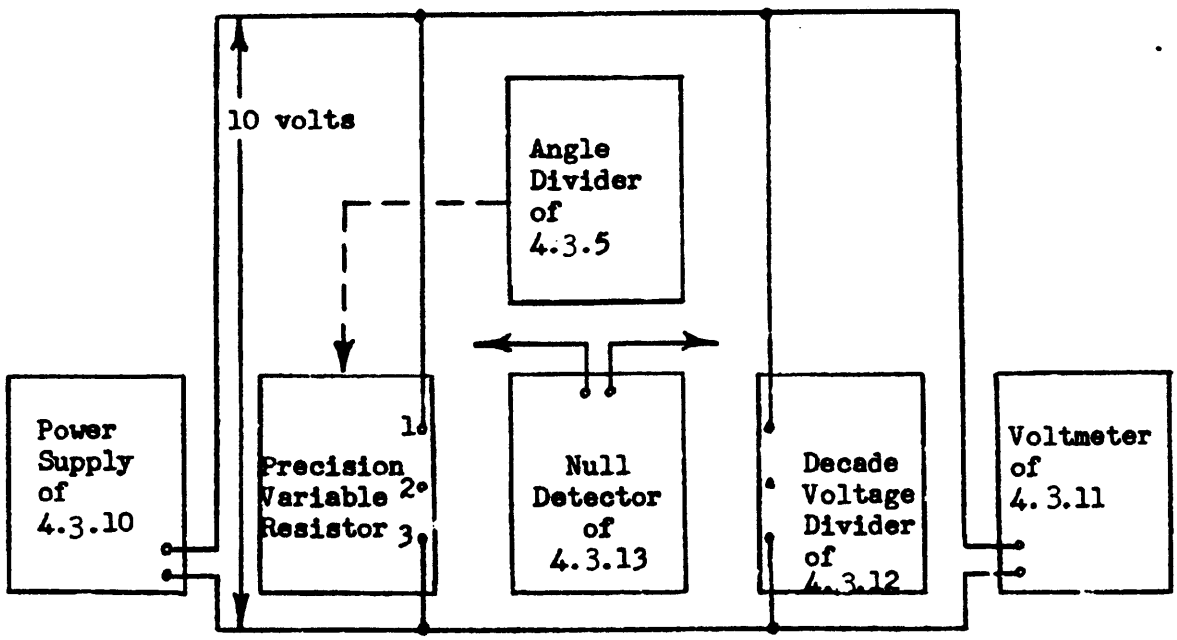


Figure 1. Test Circuit

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4.4.4.15 **Tap angle. - The** following quantities shall be measured to determine compliance with the requirements of 3.3.2.5.

(a) The angle between the tap position and the index point for Precision Variable Resistor type II,

(b) The angle between the tap position and the midpoint of the measured actual electric angle for Precision Variable Resistor type IV,

(c) The smaller of the two angles between taps for Precision Variable Resistor types V, VI, and VII.

The shaft angle **corresponding to tap position shall be determined by positioning the shaft to obtain a null indication with the Precision Variable Resists connected in accordance with figure 1, and the null detector connected between the wiper and the tap.**

The index point shall be obtained by positioning the shaft to obtain a null indication with the Precision Variable Resistor connected in accordance with figure 1, and the null detector connected between the wiper of the Precision Variable Resistor and the output terminal of the decade voltage divider with the decade voltage divider set to give the percent output voltage specified in the index information on the Precision Variable Resistor.

4.4.4.16 Tap width. - With Precision Variable Resistor types II, IV, V, VI, or VII connected in accordance with figure 1, the null detector connected between the wiper and the tap, and the shaft positioned to obtain a null indication, the greatest angle through which the shaft can be turned past the tap position in either direction without changing the reading of the null detector shall be noted to determine compliance with the requidrements of 3.3.2.6.

4.4.4.17 Electrical resolution. - With the Precision Variable Resistor connected in accordance with figure 1, and the null detector connected between the wiper of the Precision Variable Resistor and the output terminal of the decade voltage divider, the shaft shall be rotated in one direction to any position within the actual electrical angle or theoretical electrical angle, as applicable, and the decade voltage divider shall be adjusted to obtain a null indication. The shaft shall then **be** rotated slowly in the same direction through the smallest angle which will produce any indication on the null detector. This procedure shall be repeated for several different shaft positions, and the greatest incremental change in output voltage thus obtained shall be divided by the input voltage and multiplied by 100 to determine compliance with the requirements of 3.3.2.7.

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4.4.4.18 **Independent linearity. - With Precision** Variable Resis - tor **type I, IV, V, VIII, or IX connected in** accordance with figure 1, and **the null detector connected between the** wiper of the Precision Variable Resistor and the Output terminal of the decade voltage divider, the setting of the decade voltage divider required to obtain a null indication shall be recorded for equally spaced shaft positions throughout the entire actual electrical angle, starting at either end point for Precision Variable Resistor types I, IV, VIII, and IX, and at each tap for Precision Variable Resistor type V, with the increment between successive shaft angles as given below, except that the last ***increment of shaft angle*** shall be whatever increment is required to reach the other end point. **(For** this test, the two taps of Precision Variable Resistor type V shall be considered to be end points.)

Recision Variable Resistor Type	Shaft Angle Increment - Degrees
I, IV	32
V	18
VIII, Ix	90

From this data, the quantity Aj shall be computed, with shaft angle **measured** clockwise from the counterclockwise end point, where

$$\Delta j = \eta_j - \eta_1 \left(\frac{\eta_N - \eta_j}{\theta_N - \theta_1}\right)\left(\theta_j - \theta_1\right), \quad j = 1,2,\ldots,N$$

η_j = **decade voltage divider ratio for the jth measurement,**

θ_j = **shaft angle for the jth measurement (degrees),**

N = **number of measurements.**

The quantity Aj shall be plotted versus shaft angle on linear-coordinate graph paper, **a straight line shall be drawn through the points** which will minimize the maximum deviation measured parallel to the Aj axis of any point from the line, and the deviation of the point having the greatest deviation shall be multiplied by 100 to determine compliance with the requirements of 3.3.2.8. Mathematical symbols shall be interpreted in accordance with Standard MIL-STD-106. (A plot of Aj versus shaft angle shall be made for each cup of Precision Variable Resistor type V.)

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4.4.4.19 **Terminal conformity.**

4.4.4 .19.1 **Test circuit and indexing. -** With Precision Variable Resistor type II, III, VI, or VII connected in accordance with figure 1_2 and the null detector connected between the wiper of the ***Precision*** Variable Resistor and the output terminal of the decade voltage divider, the setting of the decade voltage divider required to obtain a null indication shall be recorded for equally spaced shaft positions throughout the entire theoretical electrical angle, starting at either limit of theoretical electrical angle with the increment between successive shaft angles and the limits of theoretical electrical angle as specified below.

Precision Variable Resistor Type	Shaft Angle Increment - Degrees
II	12
III	32
VI, VII	18

The shaft positions corresponding to the limits of theoretical electrical angle shall be obtained by measuring from the index point- See 4.4.4.15:

(a) For Precision Variable Resistor types II and 111, the theoretical electrical angle specified on the applicable BUWEPS drawing plus the indexing information on the Precision Variable Resistor giving the shaft angle represented by the index point defines the limits of theoretical electrical angle,

(b) For Precision Variable Resistor types VI and VII, the limits of theoretical electrical angle shall be taken as 90 degrees counterclockwise and clockwise from the index point.

4.4.4.19.2 Precision Variable Resistor type II. -For Precision Variable Resistor type IS, the quantity Aj shall be computed (for each of the two winding segments separated by the index point), with shaft angle measured in either direction from the index point, where

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$$\Delta_j = \tan\left(\frac{\theta_j}{2}\right) - \eta_j , \qquad j = 1,2,\ldots,N$$

η_j = decade voltage divider ratio for the jth measurement,

θ_j = shaft angle for the jth measurement (degrees),

N = number of measurements for each of the two segments of the winding.

The greatest Aj shall be multiplied by 100 to determine compliance with the requirements of 3.3.2.9. Mathematical symbols shall be interpreted in accordance with Standard MIL-STD-106.

4.4.4.19.3 Precision Variable Resistor type III. For Precision Variable Resistor type III, the quantity Aj shall be computed (for each cup), with shaft angle measured clockwise from the counterclockwise limit of theoretical electrical angle, where

$$\Delta_j = \frac{\theta_j - \theta_1}{320} - \eta_j , \qquad j = 1,2,\ldots,N$$

η_j = decade voltage divider ratio for the jth measurement,

θ_j = shaft angle for the jth measurement (degrees),

N = number of measurements.

The greatest Aj shall be multiplied by 100 to determine compliance with the requirements of 3.3.2.9. Mathematical symbols shall be interpreted in accordance with Standard MIL-STD-106.

4.4.4.19.4 Precision Variable Resistor types VI and VII. - For Precision Variable Resistor types VI and VII, the quantity Aj shall be computed for each of the two winding segments separated by the limits of theoretical electrical angle determined in accordance with 4.4.4.19.l(b), with shaft angle maasured in either direction from the limit of theoretical electrical thus determined nearest input terminal number 3, where

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$$\Delta_j = \frac{\theta_j}{180} - \eta_j \ , \quad j = 1,2,\ldots,N$$

η_j = decade voltage divider ratio for the jth measurement,

θ_j = shaft angle for the jth measurement (degrees),

N = number of measurements for each of the two segments of the winding.

The greatest Aj shall be multiplied by 100 to determine compliance with the requirements of 3.3.2.9. Mathematical symbols shall be Interpreted in accordance with Standard MIL-STD-106. (For Precision Variable Resistor type VII, Aj shall be computed for each cup.)

4.4.4.20 **Noise.** - With the Precision Variable Resistor connected as shown in figure 2, the shaft shall be rotated at approximately one revolution per minute in either direction, anywhere within the actual theoretical angle or theoretical electrical-angle, as applicable, and the maximum. peak-to-peak noise voltage shall be observed to determine compliance with the requirements of 3.3.2.10.

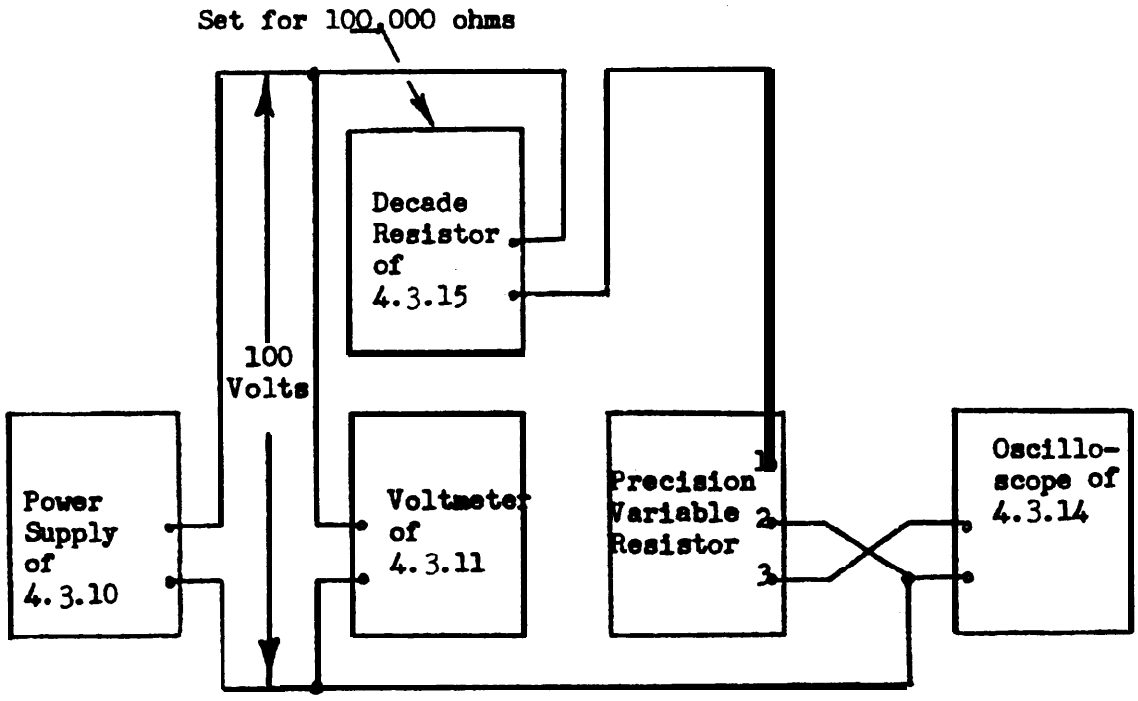


Figure 2. Circuit for Noise Test

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4.4.5 Reservation, packaging and packing. - Reservation, pack-*aging* **and** packing of the Precision Variable Resistor-shall conform to Section 5 of this specification.

4.5 Pre-production and periodic production sample tests.

4.5.1 Vibration, operating condition. - The Recision Variable Re-sister, rigidly mounted in a suitable vibration fixture, shall be sub-jected to vibration tests so as to meet the requirements specified in 3.4.1, and shall meet all the performance requirements and product characteristics **of this** specification during and after the vibration tests.

4.5.2 Vibration-transportation. - The Precision Variable Resistor, packaged and packed as specified in Section *5,* **shall be vibration tested as specified in Specification MIL-W-21927.**

4.5.3 Shock, non-operating condition. - The Precision Variable Resistor, rigidly mounted in a suitable shock fixture, shall be sub-jected to shock tests so as to meet the requirements specified in 3.4.3, and **thereafter shall meet all the performance requirements and product characteristics of this specification.**

4.5.4 Shock-transportation, storage **and handling. - The Precision Variable Resistor, packaged and packed as specified in Section 5, shall be subjected to the appropriate rough handling tests specified in Speci-fication MIL-W-21927.**

4.5.5 Temperature, operating **condition. - The Recision Variable Resistor shall be subjected to temperature tests so as to meet** the re-**quirements specified in** 3.4.5, and thereafter shall meet all the per-formance requirements and product characteristics of this specification.

4.5.6 Temperature, non-operating condition. - The Recision Vari-able Resistor shall be subjected to **temperature tests so as to meet the** requirements specified in 3.4.6, **and thereafter shall meet all the per-formance requirements and product characteristics of this specification.**

4.5.7 Humidity. - The Precision Variable Resistor shall be sub-jected to humidity tests so as to meet the requirements specified in 3.4.7, and thereafter shall meet all the performance requirements and product **characteristics of this specification.**

4.5.8 **Tropical environment. - The Precision Variable Resistor shall be exposed to fungus** tests so as to meet the requirements specified in 3.4.8, **and** thereafter shall meet all the performance requirements **and** product characteristics **of** this specification.

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4.5.9 Operating life. - The Precision Variable Resistor shall be subjected to cycling as follows so as to meet the requirements specified in 3.4.9, and thereafter shall meet all the performance requirements and product characteristics of this specification.

The shaft shall be oscillated at the rate of approximately 6 degrees per **second between the limits given in table I. The shaft position about which the oscillation takes place shall be:**

(a) The midpoint of actual electrical angle for Precision Variable Resistor types I, IV, VIII, and IX,

(b) The index point for Precision Variable Resistor type 11, 111, VI, and VII.

(c) The shaft **position 90 degrees clockwise from the position at which the wiper contacts input terminal number 3 for Recision Variable Resistor type V.**

During oscillation, the Precision Variable Resistor shall be energized in accordance with table I and figure 3, with **power alternately on for 1-1/2 hours and off for 1/2 hour. "**

4.5.9.1 100,000 cycles. - The **Precision Variable Resistor shall be subjected to 100,000 cycles of operation as described above, and starting** torque **and** electrical noise shall be measured (see *4.4.4.6* **and** *4.4.4.2o),* for the wiper within the limits of shaft oscillation, after 25,000, 50,000, 75,000, and 100,000 cycles to determine compliance with the starting torque and electrical noise requirements of 3.4.9.1.

4.5.9.2 200,000 cycles. -The Recision Variable Resistor shall be subjected to 100,000 additional cycles of operation as described above, and starting torque, independent linearity or terminal conformity - as applicable, and electrical noise shall be measured (see 4.4.4.6, 4.4.4.18, 4.4.4.19, and 4.4.4.2o), for the wiper within the limits of shaft oscillation, after 125,000, 150,000 175,000 and 200,000 cycles to determine compliance with the starting torque, linearity, and electrical noise requirements of 3.4.9.2.

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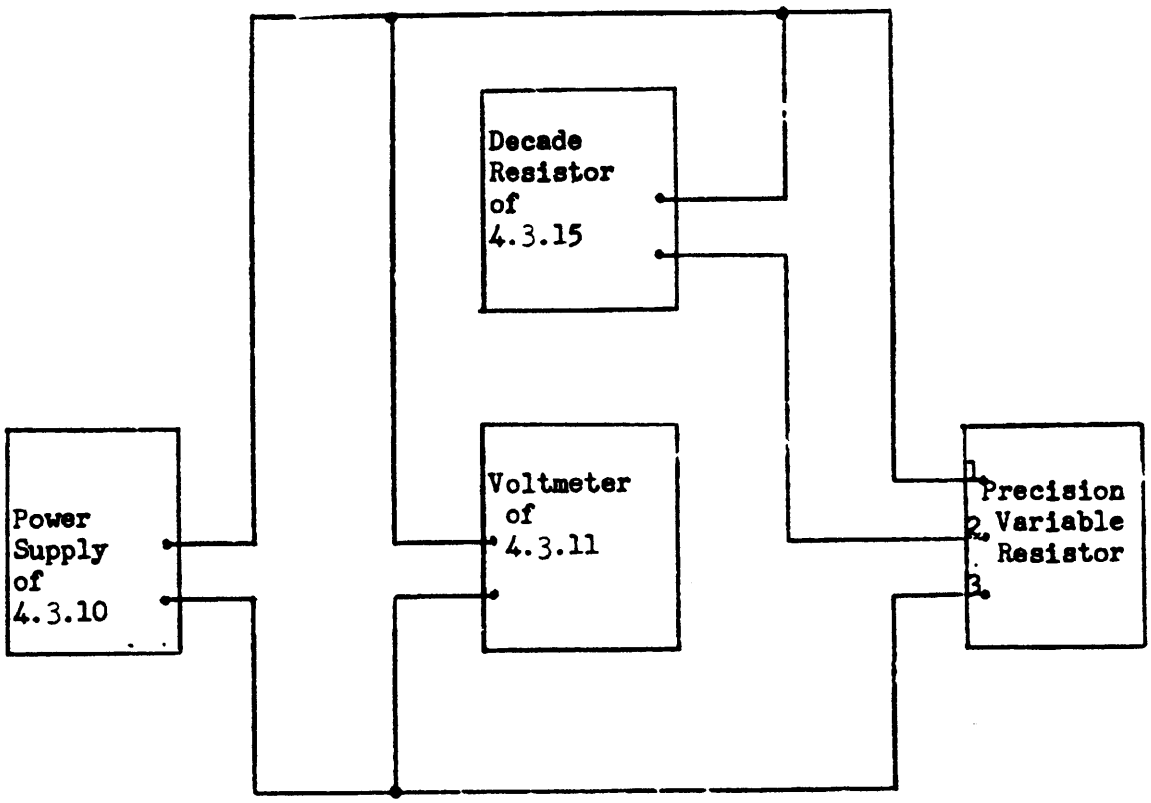


Figure 3. Circuit for Life Test

4.6 Failure of the Precision Variable Resistor to meet any of' the requirements and tests of this specification shall be considered cause for rejection.

4.7 Functional tests. - At the conclusion of the tests specified in 4.4 and *4.5,* functional and other tests, as applicable, shall be conducted on the Precision Variable Resistor to determine freedom from operational malfunction and compliance with all the requirements of section *3.*

4.8 Waiver of tests. - Tests specified in 4.4 and 4.5 maybe waived by the procuring agency for Level C packs only, provided the contractor guarantees safe delivery to the using activity.

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Table I. Life Test Conditions

Precision Variable Resistor Type	Limits of Shaft Oscillation - Degrees	Power Supply Voltage - Volts	Decade Resistor Setting - Ohms
I	± 15	28	13,375
II	± 15	32	32,000
III	± 15	5	2,475
IV	± 15	63	29,000
V	± 15	10	9,000
VI	± 15	9	4,250
VII	± 15	15	7,000
VIII	± 180	20	9,750
IX	± 180	63	29,000

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 Level A. - In addition to the following, Level A packaging shall be in accordance with Specification MIL-P-17555.

5.1.1.1 Cleaning, drying and preservative application. - Not applicable.

5.1.1.2 Unit packaging. - Unit packaging of the resistor shall be in accordance with Method IA-8 of Specification MIL-P-116. The unit package shall contain one resistor.

5.1.1.3 Intermediate package. - Resistors packaged as described in 5.1.1.2 shall be intermediate packaged in fiberboard boxes in accordance with Specification PPP-B-636, Class 2 (gross weight not to exceed 20 lbs.).

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5.1.1.4 **Cushioning. - The unit package, intermediate package and pack, shall incorporate sufficient cushioning material, bracing, or other adequate shock absorbing devices to ensure that the** resistors will **meet all the requirements of Section** 3 **after being tested in accordance with 4.5.2 and 4.5.4.**

At the time of packaging or packing, the moisture content of the cushioning material shall be no more than 5 percent. The pH range of the material shall be between 6.o and 8.o.

5.1.2 level B. - Not applicable.

5.1.3 Level c.

5.1.3.1 Cleaning, drying and **preservative application. - Not applicable.**

5.1.3.2 **Unit packaging - Unit packaging of the resistors shall be in accordance with Method III of Specification MIL-P-116.**

5.1.3.3 Intermediate package. - Intermediate packages maybe used.

5.2 Packing.

5.2.1 Level **A. - In addition to the following, Level A packing shall be in accordance with Specification MIL-p-17555.**

5.2.1.1 Exterior containers. - Resistors packaged as described in 5.1.1.3 shall be packed in a wood box in accordance with Specification PPP-B-601, overseas type, or PPP-B-621, class 2 (gross weight not to exceed 150 lbs.).

5.2.2 Level B. - In addition to the following, Level B packing shall be in accordance with Specification MIL-P-17555.

5.2.2.1 Exterior containers. - Resistors packaged as described in 5.1.1.3 shall be packed in a wood box in accordance with Specification PPP-B-601, domestic type, or PPP-B-621, Class 1 (gross weight not to exceed 150 lbs.) or in fiber boxes in accordance with Specification PPP-B-636, Class 1 (gross weight not to exceed 701bs.).

5.2.3 IevelC. - Resistors packaged in accordance with 5.1.3.2 shall be packed to afford protection against damage during shipment from the supply source to the first receiving activity for immediate use. Containers shall comply with Consolidated Freight Classification Rules or other common carrier regulations applicable to the mode of' transportation.

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5.3 **Marking.**

5.3.1 special markings. - None, unless otherwise **specified.**

5.3.2 Normal markings. - In addition to the markings required by contract or order, unit packages, intermediate packages, and shipping containers shall be marked in accordance with the requirements of MIL-STD-129 including serial numbers.

6. NOTES

6.1 **intended use. - The precision is ion Variable Resistor is intended for use in the Angle Solvers Mark 18 Mod 2 and Mod** 3, **the** Fire **Control System Mar 112 Mod 1, the Sonar Receiver Sets AN/BQG-l, -2,** -3, -4$ **or** Other suitable application.

6.2 **Ordering data. - Procurement documents should specify the specific title, number, and date of this specification and exceptions to this specification; applicable drawings and other documents; selections of applicable levels of packaging and packing; number.of intermediate** packages per pack; **number of unit containers per pack; and guarantee clause (see 6.4).**

6.2.1 The **attention of the contracting officer is invited to the Quality Assurance Provisions and options in Specification MIL-P-116 and to par 4.8.**

6.3 **Criteria for use of proper level of preservation,** packaging **and packing shall be as follows:**

For Level **A. - This level shall be used** for those items Which are to be shipped to indeterminate destination or stored under indeterminate conditions for redistribution anywhere*

For Level B. - This level shall be used only when it is definitely known that the item will be held in covered storage in overseas locations for 6 months or less or in domestic locations for an indefinite period.

For Level C. - This level shall be used only when it is definitely known that the packaged item is to be shipped to doomestic installations for immediate use at the first receiving activit y.

6.4 Guarantee. - The contract or order shall include a requirement for a guarantee stating that any Preision Vsble Resistor which fails to meet the requirments of this specification within one year after delivery because of a defect in material or workmanship or because of the vendor's failure to comply with the requerements of the applicable drawings and specification shall be replaced by the vendor at no cost to the Government.

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6.5 **The contract or order shall include a sufficient number of extra items to cover production and pre -product ion test* samples.**

NOTICE : When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government **thereby incurs no responsibility nor any obligation whatsoever; and the fact the Government may have formulated, furnished, or In anY way supplied the said drawings, specification, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented inventions that may in any way be related thereto.**

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SPECIFICATION ANALYSIS SHEET
NAVMEPS FORM 4121/8 (8-62)

FORM APPROVED
BUDGET BUREAU NO. 48-R309

INSTRUCTIONS

This sheet is to be filled out by Government or contractor personnel involved in the use of this specification in procurement of products for ultimate use by the activity shown on the reverse of this sheet.

This sheet is provided for obtaining information which will aid the activity shown in insuring that through use of this specification suitable products can be procured with a minimum amount of delay and at the least cost.

Comments and the return of this sheet will be appreciated.

Fold on dotted lines on reverse side, staple in corner, and send to the activity shown.

SPECIFICATION *(No. and abbrev. title)*

ORGANIZATION	CITY	STATE

CONTRACT NUMBER	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A DIRECT GOVERNMENT CONTRACT ☐ OR A SUBCONTRACT ☐

1. HAS ANY PART OF THIS SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
(If so, give paragraph number and wording, and recommendations for correcting the deficiencies.)

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
☐ YES ☐ NO IF YES, IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If additional papers are attached, send this form and papers in an envelope. This form is addressed for use in window envelope when appropriate.)*

SUBMITTED BY *(Print name and title)*	DATE

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DEPARTMENT OF THE NAVY
BUREAU OF NAVAL WEAPONS
WASHINGTON 25, D.C.

OFFICIAL BUSINESS

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NAVY DEPARTMENT

Chief, Bureau of Naval Weapons (RREN-5)

Navy Department

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